

Lecture 5: Solving Linear Equations and Null Space

MATH203 Applied Linear Algebra

Motivation: Why We Need to Solve Equations

Recall: The Two Languages (Lecture 4)

Every subspace has **two complementary descriptions**:

Language	How it works	Example
Descriptive	List equations (constraints)	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
Constructive	Provide recipe (generate members)	$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$

Today's question: What is each language **good at**? What is each language **bad at**?

Each Language Has Strengths and Weaknesses

	Descriptive $\{x : Ax = 0\}$	Constructive $\text{span}\{v_1, \dots\}$
Good at	Verification (check membership)	Generation (produce members)
Bad at	Generation (find examples)	Verification (check membership)

Key insight: Solving equations helps both languages do what they're bad at!

Let's see two concrete examples using Shinchon's coffee shop...

Example Setup: Shinchin's Coffee Shop Menu

Shinchan has three drinks on his menu:

	 milk	 coffee	 tea
 tea leaves	0	0	2
 lemon	0	0	1
 beans	0	2	4
 milk	1	0	1

Matrix form:

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

Case 1: Column Space (Naturally Constructive)

The column space is **naturally constructive**:

$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \right\}$$

These are the ingredient lists for , , .

In coffee shop language: $\text{Col}(A)$ = all possible ingredient combinations we can produce from our menu.

Case 1: Generation is Easy

Generation (easy): Want a vector in $\text{Col}(A)$?

Just pick a combination of drinks!

$$2 \text{ 🥛 } + 3 \text{ ☕ } = 2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 6 \\ 2 \end{pmatrix}$$

In coffee shop language: "Make 2 glasses of milk and 3 cups of coffee" produces an ingredient combination in $\text{Col}(A)$.

Generation = **making products**. Easy with constructive description!

Case 1: Verification is Hard

Verification (hard): A customer brings an ingredient list $\mathbf{b} = \begin{pmatrix} 4 \\ 2 \\ 10 \\ 3 \end{pmatrix}$.

Question in coffee shop language: Can we make a drink combination that uses exactly these ingredients?

This is asking: Is $\mathbf{b} \in \text{Col}(A)$?

Can't tell by looking! Need to **solve**: $A\mathbf{x} = \mathbf{b}$

- If solution exists $\Rightarrow$ Yes, we can make it! ($\mathbf{b} \in \text{Col}(A)$)
- If no solution $\Rightarrow$ No, impossible with our menu ($\mathbf{b} \notin \text{Col}(A)$)

Solving equations gives constructive description the power to verify!

Case 2: Null Space (Naturally Descriptive)

The null space is **naturally descriptive**:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^3 : A\mathbf{x} = \mathbf{0}\}$$

In coffee shop language: $\text{Null}(A)$ = records of relationships between drinks.

For example: If  can be made from $2 \times$ , then the relationship is recorded as:

$$\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} : \quad 2 \text{ coffees can make 1 tea}$$

Geometric meaning: $\text{Null}(A)$ stores all relationships between the columns.

Case 2: Verification is Easy (Coffee Shop Example)

Verification (easy): Someone claims "🍵 can be made from 1 🥛 and 1 ☕."

Is this relationship valid? Check if $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \in \text{Null}(A)$:

Compute:

$$\begin{aligned} 1 \text{ 🥛} + 1 \text{ ☕} - 1 \text{ 🍵} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -2 \\ -1 \\ -2 \\ 0 \end{pmatrix} \neq \mathbf{0} \end{aligned}$$

Case 2: Generation is Hard

Generation (hard): Looking at the menu, can you find the relationships?

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Question in coffee shop language: Which drinks can be made from others? What are the relationships?

Hard to see by staring! Need to **solve** $Ax = 0$.

Case 2: Solving Reveals the Relationship

After solving $A\mathbf{x} = \mathbf{0}$ (we'll learn how soon):

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \right\}$$

What does this tell us in coffee shop language?

$$\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} : \text{ 🍵 can be made from 2 🍵 }$$

Aha! This is the **relationship**: 1 cup of 🍵 uses exactly the same ingredients as 2 cups of 🍵.

Solving equations gives descriptive description the power to

Summary: The Power of Solving Equations

Subspace	Weakness	Solving provides
$\text{Col}(A)$ (constructive)	Verification "Can we make this?"	Solve $A\mathbf{x} = \mathbf{b}$ to check if possible
$\text{Null}(A)$ (descriptive)	Generation "What relationships?"	Solve $A\mathbf{x} = \mathbf{0}$ to find relationships

Today's focus: Develop a **cross-filling method** to solve $A\mathbf{x} = \mathbf{b}$, which:

- Finds all relationships between products (basis for $\text{Null}(A)$)
- Checks if a product is possible ($\mathbf{b} \in \text{Col}(A)$?)
- Determines uniqueness (Is there only one way?)

Null Space: The Relationship Catalog

Definition: Null Space

Definition (Null Space (Kernel))

For a matrix $A \in \mathbb{R}^{m \times n}$, the **null space** is:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Meaning: $\text{Null}(A)$ records all **relationships between the columns** of A .

Also called the **kernel**, denoted $\ker(A)$.

In coffee shop language: Drink combinations that reveal "which drinks can be made from others."

Immediate Example: Is This a Valid Relationship?

From Shinchin's menu (A from before), someone claims:

" can be made from 2 "

This means: $\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ should be in $\text{Null}(A)$.

Let's verify:

$$A \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ -1 \end{pmatrix} \neq \mathbf{0}$$

Result: This is NOT a valid relationship! (Actually  $\neq 2 \times$ )

What Would Be a Valid Relationship?

Let's try another: someone claims "  can be made from 2  plus 1  "

This means: $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ might be in $\text{Null}(A)$.

$$A \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ 0 \end{pmatrix} \neq \mathbf{0}$$

Still not valid!

How do we **systematically find** valid relationships? We need to solve $A\mathbf{x} = \mathbf{0}$!

Key Insight: Null Space Records Relationships

Important:

- Each vector $\mathbf{x} \in \text{Null}(A)$ records a relationship
- $\text{Null}(A) = \text{"relationship catalog"}$ between columns

Size of the catalog:

- $\text{Null}(A) = \{\mathbf{0}\} \rightarrow$ no relationships $\rightarrow$ columns **independent**
- $\dim(\text{Null}(A)) = k \rightarrow$ there are k independent relationships

Null Space is a Subspace

Proposition

$\text{Null}(A)$ is a subspace of $\mathbb{R}^n$.

Proof: Verify the three axioms:

1. $\mathbf{0} \in \text{Null}(A)$ since $A\mathbf{0} = \mathbf{0}$ ✓
2. If $A\mathbf{x}_1 = \mathbf{0}$ and $A\mathbf{x}_2 = \mathbf{0}$, then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = \mathbf{0}$$

✓

3. If $A\mathbf{x} = \mathbf{0}$, then $A(c\mathbf{x}) = \mathbf{0}$ ✓

The Problem: We Need Constructive Description

Descriptive: $\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$

- **Verification** (easy): Is $\mathbf{v}$ a valid relationship? Compute $A\mathbf{v}$
- **Generation** (hard): Find all relationships? Can't guess!

Goal: Convert to **constructive**:

$$\text{Null}(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

Method: Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling!

But first — to solve $A\mathbf{x} = \mathbf{b}$ in general, we need augmented matrices...

From $Ax = b$ to Augmented Matrix

Motivation: Why Augmented Matrix?

We want to solve: $A\mathbf{x} = \mathbf{b}$

Column view:

$$A\mathbf{x} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

We're looking for **coefficients** $x_1, \dots, x_n$ that combine columns to get $\mathbf{b}$.

Key idea: Add $\mathbf{b}$ as an extra column...

The Augmented Matrix

Form the **augmented matrix**:

$$(A \mid \mathbf{b}) = \left(\begin{array}{c|c|c|c|c} | & | & & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \\ | & | & & | & | \end{array} \right)$$

This is an $m \times (n + 1)$ matrix with $n + 1$ columns.

Now: What happens if we take a linear combination with coefficients

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix} ?$$

The Key Connection

Take $(A \mid \mathbf{b})$ and multiply by $\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix}$:

$$(A \mid \mathbf{b}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n + (-1)\mathbf{b}$$

This equals $\mathbf{0}$ if and only if:

$$x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

which is exactly $A\mathbf{x} = \mathbf{b}$!

Why This Matters

Crucial observation:

$A\mathbf{x} = \mathbf{b}$ has solution $\iff \begin{pmatrix} \mathbf{x} \\ -1 \end{pmatrix}$ records a relationship in $(A \mid \mathbf{b})$

But we won't compute $\text{Null}(A \mid \mathbf{b})$ directly. Instead:

We'll use **cross-filling** on $(A \mid \mathbf{b})$ to extract simplified equations, then solve backwards!

The Cross-Filling Method for Solving $Ax = b$

Recall: Cross-Filling Decomposition (Lecture 3)

Any matrix can be decomposed via cross-filling:

$$A = R_1 + R_2 + \cdots + R_r$$

where each R_i is a **rank-one matrix**.

Key property: All rows of a rank-one matrix are scalar multiples of each other.

Example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

Apply Cross-Filling to Augmented Matrix

For $(A \mid \mathbf{b})$, we cross-fill:

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2) + \cdots + (R_r \mid \mathbf{b}_r)$$

Key observation: Since $(R_i \mid \mathbf{b}_i)$ is rank-one, all its rows are proportional — they represent the **same equation!**

One rank-one piece = one independent equation.

Why Are These Equations Valid?

Question: Why is solving these r equations equivalent to the original m equations?

Answer: Cross-filling gives $(A \mid \mathbf{b}) = UV$ where:

- U is $m \times r$ with **linearly independent columns**
- V is $r \times (n + 1)$

Since U 's columns are independent:

$$UV\mathbf{z} = \mathbf{0} \iff UV\mathbf{z} = U\mathbf{0} \iff V\mathbf{z} = \mathbf{0}$$

(Left multiplication by independent columns can be cancelled.)

What is V Explicitly?

$$V = \begin{pmatrix} \text{— row from } (R_1 \mid \mathbf{b}_1) \text{ —} \\ \text{— row from } (R_2 \mid \mathbf{b}_2) \text{ —} \\ \vdots \\ \text{— row from } (R_r \mid \mathbf{b}_r) \text{ —} \end{pmatrix}$$

V is an $r \times (n + 1)$ **augmented matrix**, where:

- Each row is one simplified equation
- The last column corresponds to the right-hand side

In practice: We'll see V explicitly in our example!

Observation: Variables Decrease

As cross-filling proceeds, each piece has fewer active variables:

- $(R_1 \mid \mathbf{b}_1)$: involves **all n variables**
- $(R_2 \mid \mathbf{b}_2)$: involves **$n - 1$ variables** (column j_1 eliminated)
- $(R_3 \mid \mathbf{b}_3)$: involves **$n - 2$ variables**
- $\vdots$
- $(R_r \mid \mathbf{b}_r)$: involves **fewest variables**

Strategy: Start from the end (fewest variables) and work backwards!

Concrete Example: Step-by-Step Visual Cross-Filling

Example Setup

Solve:

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 11 \\ 16 \end{pmatrix}$$

Form augmented matrix $(A \mid \mathbf{b})$ and cross-fill visually!

Step 1: Initial Augmented Matrix

$$\left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 1 & 2 & 11 \\ 3 & 6 & 1 & 3 & 16 \end{array} \right)$$

Goal: Use cross-filling to decompose into rank-one pieces.

Strategy: Choose pivots to eliminate columns one by one.

Step 2: Choose First Pivot at (1,1)

$$\text{column} \begin{pmatrix} \overset{\text{center}}{\uparrow} \\ \leftarrow 1 \rightarrow 2 \rightarrow 0 \rightarrow 1 \rightarrow 5 \rightarrow \\ 2 \quad 4 \quad 1 \quad 2 \quad 11 \\ \downarrow \\ 3 \quad 6 \quad 1 \quad 3 \quad 16 \end{pmatrix}$$

Pivot = entry $(1,1) = 1$ (highlighted red).

Extract: Column $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \text{Row } (1 \quad 2 \quad 0 \quad 1 \quad | \quad 5)$

Step 3: First Rank-One Piece R_1

$$R_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \end{array} \right)$$

Expanding:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 0 & 2 & 10 \\ 3 & 6 & 0 & 3 & 15 \end{array} \right) \leftarrow R_1$$

Equation from R_1 (read any row, e.g., row 1):

$$x_1 + 2x_2 + 0x_3 + 1x_4 = 5$$

Step 4: Compute Remainder

Before: $(A \mid \mathbf{b})$

$$\left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 1 & 2 & 11 \\ 3 & 6 & 1 & 3 & 16 \end{array} \right) - \left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 0 & 2 & 10 \\ 3 & 6 & 0 & 3 & 15 \end{array} \right)$$

After: Remainder

$$\left(\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Column 1 eliminated! Next pivot at $(2, 3)$ (blue).

Step 5: Choose Second Pivot at (2,3)

$$\left(\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Pivot = entry $(2, 3) = 1$ (blue).

Extract: Column $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \text{Row } (0 \ 0 \ 1 \ 0 \mid 1)$

Step 6: Second Rank-One Piece R_2

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 & | & 1 \end{pmatrix}$$

Expanding:

$$\left(\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right) \leftarrow R_2$$

Equation from R_2 (read any non-zero row):

$$x_3 = 1$$

Step 7: Final Remainder is Zero

After subtracting R_2 :

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Cross-filling complete!

$$(A \mid \mathbf{b}) = R_1 + R_2$$

Step 8: The Matrix V

Recall: $(A \mid \mathbf{b}) = UV$ where rows of V come from each R_i .

Explicitly:

$$V = \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

- Row 1 from R_1 : $x_1 + 2x_2 + 0x_3 + x_4 = 5$
- Row 2 from R_2 : $0x_1 + 0x_2 + x_3 + 0x_4 = 1$

Solving $(A \mid \mathbf{b})$ is equivalent to solving V !

Step 9: How to Identify Pivot and Free Variables

From R_2 to R_1 , we went from 1 active variable to 4 active variables.

Which new variables appeared? x_1, x_2, x_4 (three new variables)

Which one did we choose as pivot in R_1 ? x_1 (at position (1,1))

Therefore:

- $x_1 =$ **pivot variable** (we chose it)
- $x_2, x_4 =$ **free variables** (appeared but not chosen)

General rule: From R_{i+1} to R_i , new variables appear. The one we chose as pivot = pivot variable. Others = free variables.

Step 10: Summary of Variables

Pivot columns: 1, 3

- **Pivot variables:** x_1, x_3 (determined by equations)

Non-pivot columns: 2, 4

- **Free variables:** x_2, x_4 (can choose any value)

Existence check: No equation of form $0 = c$ ($c \neq 0$) found.

$\Rightarrow$ **Solution exists!**

Step 11: Backward Solving

From R_2 (simplest equation):

$$x_3 = 1$$

From R_1 :

$$x_1 + 2x_2 + 0x_3 + x_4 = 5$$

Substituting $x_3 = 1$:

$$x_1 + 2x_2 + 0(1) + x_4 = 5$$

$$x_1 + 2x_2 + x_4 = 5$$

Solve for pivot variable x_1 :

Step 12: Write General Solution

Let free variables $x_2 = s$ and $x_4 = t$ (parameters).

General solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 - 2s - t \\ s \\ 1 \\ t \end{pmatrix}$$

Decompose:

$$= \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Step 13: Interpret the Solution

$$\mathbf{x} = \underbrace{\begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{\text{Particular solution } \mathbf{x}_p} + s \underbrace{\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{\text{Null vector } \mathbf{v}_1} + t \underbrace{\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\text{Null vector } \mathbf{v}_2}$$

Particular solution ($s = t = 0$): $\mathbf{x}_p = \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

Null space basis:

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Applications: Existence, Uniqueness, Null Space

Application 1: Finding Null Space Basis

Proposition

To find a basis for $\text{Null}(A)$:

1. Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling
2. Identify free variables
3. For each free variable, set it = 1, others = 0
4. Solve for pivot variables

Result: $\text{Null}(A) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$

Descriptive $\rightarrow$ Constructive conversion complete!

Application 1: Concrete Example

From our example, free variables are x_2, x_4 .

For $\mathbf{v}_1$: Set $x_2 = 1, x_4 = 0$

$$x_3 = 1 \quad (\text{from } R_2)$$

$$x_1 = 5 - 2(1) - 0 = 3 \quad \text{wait, this gives particular solution!}$$

Correction: For null space, solve $A\mathbf{x} = \mathbf{0}$ (not $\mathbf{b}$)!

From R_1 with $\mathbf{b} = \mathbf{0}$: $x_1 + 2x_2 + x_4 = 0$

$$x_1 = -2x_2 - x_4$$

$$\text{Set } x_2 = 1, x_4 = 0: \mathbf{v}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Application 2: Existence Criterion — The Key Insight

Question: When does $A\mathbf{x} = \mathbf{b}$ have a solution?

Strategy: Cross-fill $(A \mid \mathbf{b})$ using pivots from A only.

- Use $r = \text{rank}(A)$ pivots to eliminate A part
- After r steps, all pivot rows are cleared
- Then check: **What's left in the non-pivot rows?**

Let's see this visually with colored rows. . .

Case 1: All Equations Effective (Solution Exists)

After using r pivots from A :

$$\begin{array}{l} \text{pivot row 1} \\ \text{pivot row 2} \\ \text{non-pivot row} \end{array} \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Observation:

- Pivot rows (colored): cleared by design
- Non-pivot rows: **passively became 0** (including **b** part)

Conclusion: No equation of form $0 = c$ ($c \neq 0$).

All equations are **effective** (not contradictory).

Solution exists!

Case 2: Contradiction Appears (No Solution)

After using r pivots from A :

$$\begin{array}{l} \text{pivot row 1} \\ \text{pivot row 2} \\ \text{non-pivot row} \end{array} \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c \end{array} \right) c \neq 0$$

Observation:

- Pivot rows: cleared
- Non-pivot row: A part is 0, but $\mathbf{b}$ part $\neq 0$

This gives equation: $0 = c$ where $c \neq 0$ — **contradiction!**

No solution!

Existence Theorem: Correct Logic

Theorem (Existence of Solutions)

$A\mathbf{x} = \mathbf{b}$ has a solution if and only if:

$$\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Correct logic:

- The cross-filling algorithm **always proceeds**
- If $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) \Rightarrow$ no contradiction $\Rightarrow$ solution exists
- If $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) + 1 \Rightarrow$ contradiction $0 = c$ appears $\Rightarrow$ **any $\mathbf{x}$ violates this equation** $\Rightarrow$ no solution

Geometric Interpretation

$$\mathbf{b} \in \text{Col}(A) \iff \text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Adding $\mathbf{b}$ as a column doesn't increase rank $\iff \mathbf{b}$ is already in span of A 's columns.

Coffee shop: Customer's ingredient list is achievable $\iff$ it's in our column space.

Application 3: Uniqueness Criterion

Theorem (Uniqueness of Solutions)

If $Ax = \mathbf{b}$ has a solution, it is **unique** if and only if:

$$\text{Null}(A) = \{\mathbf{0}\}$$

Equivalently:

- No free variables
- All columns are pivot columns
- $\text{rank}(A) = n$ (number of columns)

Proof idea: If $\mathbf{x}_1, \mathbf{x}_2$ are both solutions, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.

Application 4: Structure of Solution Set

Theorem (General Solution Structure)

If $A\mathbf{x} = \mathbf{b}$ has solutions, **all solutions** have the form:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$$

where:

- $\mathbf{x}_p$ is any **particular solution**
- $\mathbf{x}_n \in \text{Null}(A)$ is arbitrary

Geometric picture: Solution set is $\mathbf{x}_p + \text{Null}(A)$ (affine subspace).

Dimension Formula: Rank-Nullity Theorem

Counting Pivot and Free Variables

From cross-filling an $m \times n$ matrix A :

- Number of pivot columns = $\text{rank}(A)$
- Number of free variables = $\dim(\text{Null}(A))$
- Total columns = n

Every column is either pivot or non-pivot:

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

This is the **Rank-Nullity Theorem**.

Visual Example: Rank-Nullity

From our example:

$$\begin{array}{c} \text{pivot} \quad \text{free} \quad \text{pivot} \quad \text{free} \\ \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 0 & 1 \\ \hline 2 & 4 & 1 & 2 \\ \hline 3 & 6 & 1 & 3 \\ \hline \end{array} \right) \end{array}$$

Pivot columns (rank = 2)

Non-pivot columns (nullity = 2)

$$\underbrace{2}_{\text{rank}} + \underbrace{2}_{\text{nullity}} = \underbrace{4}_n$$

Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)

For any $m \times n$ matrix A :

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

Proof idea: Every column is either:

- A pivot column (contributes to rank), or
- A non-pivot column (contributes one basis vector to null space)

These counts sum to n .

Preview: Null Space and Row Space Orthogonality

Natural Orthogonality

Theorem

$\text{Null}(A) \perp \text{Row}(A)$

Every vector in $\text{Null}(A)$ is orthogonal to every vector in $\text{Row}(A)$.

Proof: Let $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{r}_i = i\text{-th row}$.

By definition: $A\mathbf{x} = \mathbf{0}$ means $\mathbf{r}_i \cdot \mathbf{x} = 0$ for all i .

Coffee Shop Interpretation

If $A = \text{"drinks"} \rightarrow \text{ingredients}$:

- **Row space:** All ingredient combinations from drinks
- **Null space:** Relationships between drinks (which can be made from others)

Perpendicular: A relationship (which drinks are equivalent) has nothing in common with actual ingredient requirements!

Next lecture: Four fundamental subspaces and complete orthogonal structure.

Summary

What We Achieved Today

1. Null Space:

- Descriptive: $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
- Records relationships between columns
- Conversion to constructive requires solving

2. Cross-Filling Method:

- Each rank-one piece = one equation
- Matrix V explicitly shows simplified equations
- Variables decrease $\rightarrow$ backward solving

What We Achieved Today (continued)

3. Existence: $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) \iff$ no contradiction

- Visual: pivot rows cleared, non-pivot rows passive
- If non-pivot row has $0 = c$ ($c \neq 0$) $\rightarrow$ no solution

4. Uniqueness: $\text{Null}(A) = \{\mathbf{0}\} \iff$ unique solution

5. Rank-Nullity: $\text{rank}(A) + \text{nullity} = n$ (visual: colored columns)

6. Orthogonality: $\text{Null}(A) \perp \text{Row}(A)$

Next lecture: Four fundamental subspaces!